

Example 3

The Jaccard Similarity of $\{a, a, b, d\}$ and $\{a, a, b, b, b, c\}$ is $3/10$. The intersection counts a twice and b once, so its size is 3. The size of the union of two sets is always the sum of the sizes of the two sets, or 10 in this case. Since the highest possible Jaccard Similarity is $1/2$, the score of 30% is quite good.

It may be difficult to detect the similarity between movies and users because we have little information about movie-item pairs in this sparse-utility matrix. Even if two items belong to the same genre, there are likely to be very few users who bought or rated both. Similarly, even if two users both like a genre or multiple genres, they may not have bought any items to indicate that. One way of dealing with this lack of data is to cluster movies and/or users. For example, cluster similar movies based on their properties. After clustering the movies, we can revise the utility matrix so the columns represent clusters of movies and the entry for User u and Cluster c is the average rating that u gave to the members of Cluster c that u did not individually rate. Note that u may have rated none of the cluster members, in which case the entry for u and c is left blank. Other similarity measures will have to be used to identify similar items. Figure 5.6 shows the revised utility matrix for cluster users and movies

	HP	SW
Ann	4	1
Bob	4.67	
Carl		4.5
Doug	3	3

Figure 5.6 Utility matrix for Clusters.

5.7 Distance Measures

We have seen the importance of finding how similar two items are in the earlier sections. Similarity is a measure of how much alike two data objects are. In a complementary fashion, Similarity in a data-mining context is usually described as a distance measure between the two items. A small distance indicates a high degree of similarity and a large distance indicates a small degree of similarity.

A distance measure indicates the degree of dissimilarity between the two items. Distance is a subjective measure and is highly dependent on the domain and application. Thus, several different types of Distance Measures become relevant in different domains.

We shall discuss a few of the popular Distance Measures and their applications in the following section.

5.7.1 Definition of a Distance Metric

It is a numerical measure of how different two data objects are. It is a function that maps pairs of objects to real values.

1. Is lower when objects are more alike.
2. Minimum distance is 0 when comparing an object with itself.
3. Upper limit varies.

More formally, a distance function d is a distance metric if it is a function from pairs of objects to real numbers such that:

1. $d(x, y) > 0$. (Non-negativity)
2. $d(x, y) = 0$ iff $x = y$. (Identity)
3. $d(x, y) = d(y, x)$. (Symmetry)
4. $d(x, y) < d(x, z) + d(z, y)$. (Triangle inequality)

The triangle inequality property guarantees that the distance function is well-behaved. It indicates that the direct connection is the shortest distance between two points. It says, intuitively, that to travel from x to y , we cannot obtain any benefit if we are forced to travel via some particular third point z .

It is useful also for proving properties about the data. For example, suppose we want to find an object that minimizes the sum of distances to all points in a dataset; if we select the best point from the dataset, the sum of distances we get is at most twice that of the optimal point (see Fig. 5.7).

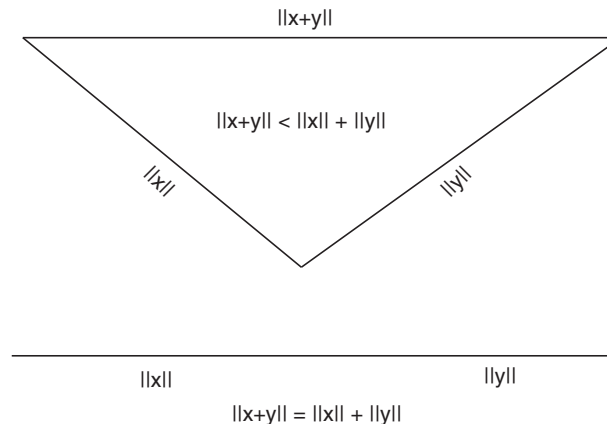


Figure 5.7 Triangle inequality illustration.

5.7.2 Euclidean Distances

The easiest distance measure to compute is called Manhattan Distance or cab-driver distance. In the 2D case, each point is represented by an (x, y) point. So consider two points (x_1, y_1) and (x_2, y_2) . The Manhattan Distance is then calculated by

$$|x_1 - x_2| + |y_1 - y_2|$$

It is called “Manhattan Distance” because it is the distance one would have to travel between points if one were constrained to travel along grid lines, as on the streets of a city such as Manhattan. Manhattan Distance measure is a special case of a distance measure in a “Euclidean Space”. An n -dimensional Euclidean Space is one where each data point is a vector of n real numbers.

The general form of distance measure used for Euclidean spaces is called L_r norm. For any constant r , we can define the L_r -norm to be the distance measure d defined by

$$d([x_1, x_2, \dots, x_n], [y_1, y_2, \dots, y_n]) = \left(\sum_{i=1}^n |x_i - y_i|^r \right)^{1/r}$$

This general form of the distance measure is called the **Minkowski measure**. Manhattan Distance is a special case of the Minkowski measure also called the **L_1 -norm**. The conventional distance measure in this space, which we shall refer to as the **L_2 -norm**, is defined as

$$d([x_1, x_2, \dots, x_n], [y_1, y_2, \dots, y_n]) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

This is also called as the **Euclidean Distance**. This measure is derived from the Pythagoras theorem of straight line distance. By taking the positive square root, we see that the distance can never be negative. When the two points are identical, we take the distance to be zero. Because of the square of the differences, the measure is symmetric, that is, $(x_i - y_i)^2 = (y_i - x_i)^2$. The Pythagoras theorem

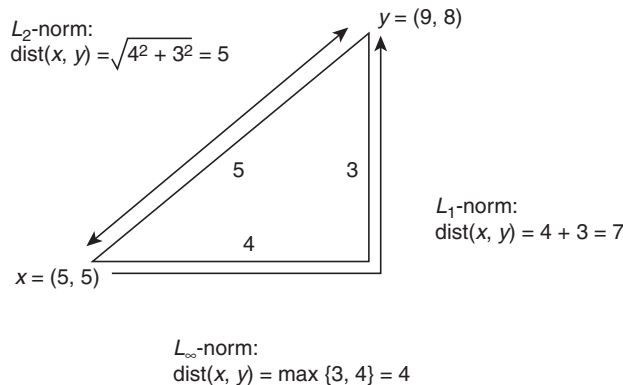


Figure 5.8

itself provides proof of the triangle inequality. Thus, we see that Euclidean Distance is a true-distance measure.

An interesting distance measure is the L_∞ -norm, which is the limit as r approaches infinity of the L_r -norm. As r gets larger, only the dimension with the largest difference matters, so formally, the L_∞ -norm is defined as the maximum of $|x_i - y_i|$ over all dimensions i . An example is shown in Fig. 5.8.

Example 4

Consider the data matrix with four points x_1, x_2, x_3, x_4 with two attributes each in Fig. 5.9. The matrices below indicate the computations of the Manhattan and Euclidean distances.

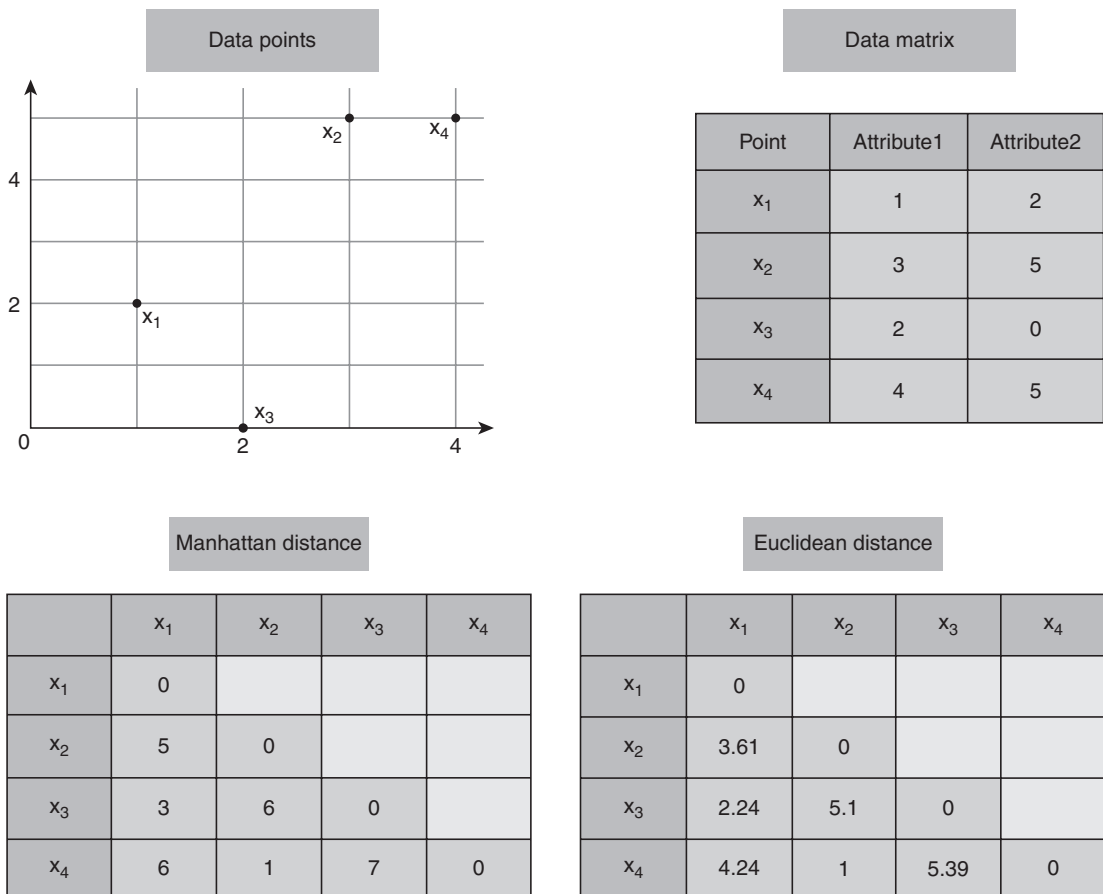


Figure 5.9

5.7.3 Jaccard Distance

The **Jaccard Distance**, which measures dissimilarity between the sample sets, is complementary to the Jaccard coefficient and is obtained by subtracting the Jaccard coefficient from 1, or equivalently, by dividing the difference of the sizes of the union and the intersection of the two sets by the size of the union:

$$d_J(A, B) = 1 - J(A, B) = \frac{|A \cup B| - |A \cap B|}{|A \cup B|}$$

Jaccard Distance has all the constraints of a distance measure:

1. Since the size of intersection is always less than or equal to the size of the union, Jaccard Distance satisfies the non-negative constraints.
2. The size of the union and the intersection of the two sets can never be same at the same time except the case when both the sets are same. Jaccard Similarity is 1 only when the same sets are used. Further, Jaccard Distance is strictly positive.
3. Union and intersection of two sets are always symmetric; Jaccard Distance satisfies symmetric axiom.
4. Jaccard Similarity always satisfies triangular inequality, and therefore, so does Jaccard Distance.

5.7.4 Cosine Distance

The Cosine Distance between two points is the angle formed between their vectors. The angle always lies in between 0° and 180° regardless of the number of dimensions. The Cosine Distance is measurable in Euclidean spaces of any dimensional spaces where vectors of points in spaces are integer or Boolean components. In this case, points can be considered as directions. Then the cosine distance between the two points is the angle that the vectors to those points make. This angle will be in the range $0^\circ - 180^\circ$, regardless of how many dimensions the space has.

We can calculate the Cosine Distance by first computing the cosine of the angle, and then applying the arc-cosine function to translate to an angle in the $0^\circ - 180^\circ$ range. The cosine of the angle between the two vectors is calculated as the dot product of vectors by dividing with the **L2**-norms of x and y . Figure 5.10 depicts the computation of the Cosine Distance between two documents D1 and D2.

Cosine Distance is a distance measure:

1. Since the values are taken in the range of $(0, 180)$, so there cannot be any negative distances.
2. Angle between the two vectors is 0 only if they are in the same direction.

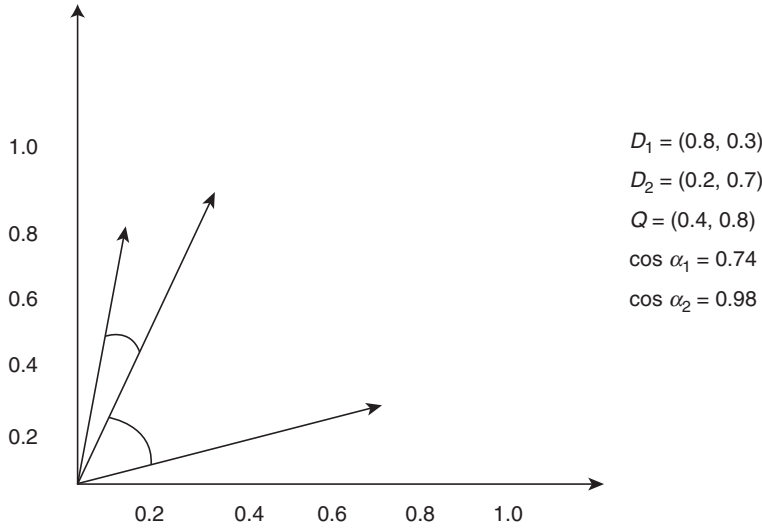


Figure 5.10 Cosine Distance between two documents.

3. Angle between the two vectors satisfies symmetry. The angle between x and y is the same as the angle between y and x .
4. Cosine Distance also satisfies triangle inequality. The triangle inequality is best argued by physical reasoning. One way to rotate from x to y is to rotate to z and then to y . The sum of those two rotations cannot be less than the rotation directly from x to y .

Example 5

Cosine Similarity

Find the similarity between documents 1 and 2.

$$D1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)$$

$$D2 = (3, 0, 2, 0, 0, 1, 1, 0, 1, 0, 1)$$

Solution

$$D1 \cdot D2 = 5 * 3 + 0 * 0 + 3 * 2 + 0 * 0 + 2 * 1 + 0 * 1 + 0 * 1 + 2 * 1 + 0 * 0 + 0 * 1 = 25$$

$$\|D1\| = (5 * 5 + 0 * 0 + 3 * 3 + 0 * 0 + 2 * 2 + 0 * 0 + 0 * 0 + 2 * 2 + 0 * 0 + 0 * 0)^{0.5} = (42)^{0.5} = 6.481$$

$$\|D2\| = (3 * 3 + 0 * 0 + 2 * 2 + 0 * 0 + 1 * 1 + 1 * 1 + 0 * 0 + 1 * 1 + 0 * 0 + 1 * 1)^{0.5} = (17)^{0.5} = 4.12$$

$$\cos(D1, D2) = 0.94$$

The angle ϕ that gives cosine value as 0.94 is the cosine distance between the two vectors.

5.7.5 Edit Distance

Edit Distance is used for the string points in vector space. Consider the two strings:

$$x = x_1 x_2 \dots x_n \quad \text{and} \quad y = y_1 y_2 \dots y_m$$

The minimum number of times the insertions and deletions of single characters is done to convert x to y or y to x is the distance between them. For example, the Edit Distance between “Hello” and “Jello” is 1. The edit distance between “good” and “goodbye” is 3. The Edit Distance between any string and itself is 0. Dynamic programming is used to compute the Edit Distance between two strings. The longest common subsequence (LCS) of x and y can also be used to calculate edit distance. An LCS of x and y is a string that is constructed by deleting positions from x and y , and that is as long as any string that can be constructed that way. The Edit Distance can be obtained by subtracting twice the length of their LCS from the sum of lengths of x and y :

$$d(x, y) = |x| + |y| - 2|\text{LCS}(x, y)|$$

Edit Distance is a distance measure because of the following reasons:

1. To convert string x into string y , number of insertions and deletions can never be negative.
2. Edit Distance between two strings is zero only if the two strings are same.
3. Number of insertions and deletions when converting string x to y or y to x is the same.
4. Number of insertions and deletions when converting string x to z plus z to y will always be greater than or equal to that of converting string x directly to y .

Example 6

Let $x = abcde$ and $y = bcduve$. Turn x into y by deleting a ; then insert u and v after d . **Edit-distance = 3.**

Now $\text{LCS}(x, y) = bcde$. So

$$|x| + |y| - 2|\text{LCS}(x, y)| = 5 + 6 - 2 * 4 = 3$$

5.7.6 Hamming Distance

The Hamming Distance is used for the Boolean vectors, that is, which contain only 0 or 1. The number of items in which the two items differ is the Hamming Distance between them. Hamming Distance is definitely a distance measure.

1. The Hamming Distance can never be negative.
2. The Hamming Distance is zero only if the vectors are identical.
3. The Hamming Distance is same irrespective of the order of the vectors and does not depend on which of two vectors we take first. So, it is symmetrical.
4. The triangle inequality is also satisfied because if x is the number of component in which p and r differ, and y is the number of component in which r and q differ, then p and q cannot differ in more than $x + y$ components.

Example 7

Consider two vectors, $p_1 = 10101$; $p_2 = 10011$. Then

$$d(p_1, p_2) = 2$$

because the bit-vectors differ in the 3rd and 4th positions.

Example 8

Consider two vectors, $x = 010101001$; $y = 010011000$. Hamming Distance = 3; there are three binary numbers different between the x and y .

Summary

- **Jaccard Similarity:** The Jaccard Similarity of sets is the ratio of the size of the intersection of the sets to the size of the union. This measure of similarity is suitable for many applications, including textual similarity of documents and similarity of buying habits of customers.
- **Distance Measures:** A Distance Measure is a function on pairs of points in a space that satisfy certain axioms. The distance between two points is 0 if the points are the same, but greater than 0 if the points are different. The distance is symmetric; it does not matter in which order we consider the two points. A distance measure must satisfy the triangle inequality: the distance between the two points is never more than the sum of the distances between those points and some third point.
- **Euclidean Distance:** The most common notion of distance is the Euclidean Distance in an n -dimensional space. This distance,

sometimes called the **L2-norm**, is the square root of the sum of the squares of the differences between the points in each dimension. Another distance suitable for Euclidean spaces, called Manhattan distance or the **L1-norm**, is the sum of the magnitudes of the differences between the points in each dimension.

- **Jaccard Distance:** One minus the Jaccard Similarity is a distance measure, called the Jaccard Distance.
- **Cosine Distance:** The angle between vectors in a vector space is the Cosine Distance measure. We can compute the cosine of that

angle by taking the dot product of the vectors and dividing by the lengths of the vectors.

- **Edit Distance:** This distance measure applies to a space of strings, and is the number of insertions and/or deletions needed to convert one string into the other. The Edit Distance can also be computed as the sum of the lengths of the strings minus twice the length of the longest common subsequence of the strings.
- **Hamming Distance:** This distance measure applies to a space of vectors. The Hamming Distance between two vectors is the number of positions in which the vectors differ.

Exercises

1. Identify five applications not discussed in this chapter for collaborative filtering.
2. What is the difference between user-based collaborative filtering and item-based collaborative filtering?
3. Find Jaccard distances between the following pairs of set:
 - (a) $\{1, 2, 3, 4\}$ and $\{2, 3, 5, 7\}$
 - (b) $\{3, 2, 1\}$ and $\{2, 4, 6\}$
 - (c) $\{1, 2, 3, 4\}$ and $\{5, 6, 7, 8\}$
4. For each of the following vectors, x and y , calculate the cosine distance measure:
 - (a) $x = (1\ 1\ 0\ 0\ 0)$, $y = (0\ 0\ 0\ 1\ 1)$
 - (b) $x = (0\ 1\ 0\ 1\ 1)$, $y = (1\ 0\ 1\ 0\ 0)$
 - (c) $x = (0\ 1\ 2\ 4\ 5\ 3)$, $y = (5\ 6\ 7\ 9\ 10\ 8)$
5. What is the range of values that are possible for the cosine measure? If two objects have a cosine measure of 1, are they identical? Explain.
6. Compute the Hamming distance between the following:
 - (a) $x = 0101010001$ and $y = 0100011000$
 - (b) $x = 111111100$ and $y = 000111111$
7. Suppose that you are comparing how similar two organisms of different species are in terms of the number of genes they share. Describe which measure, Hamming or Jaccard, you think would be more appropriate for comparing the genetic makeup of two organisms. Explain. (Assume that each animal is represented as a binary vector, where each attribute is 1 if a particular gene is present in the organism and 0 otherwise.)
8. Find edit distance between the following pair of strings:

- (a) “abcdef” and “bcdesg”
- (b) “abcceghi” and “defghsi”
- (c) “aaaaaabc” and “bbbbbbef”

- 9. You have studied various similarity and distance measures in this chapter. Give two example applications where the use of a particular distance measure is most appropriate. Explain your answer.

Programming Assignments

- 1. Consider very large vectors indicating n -dimensional data points. Write programs using the MapReduce framework for computing the following distance measures for any two large vectors. {Assume data to be having more than 1000 dimensions}
 - (a) Jaccard Distance
 - (b) Cosine Distance
- 2. Now use the above written codes for distance measures to perform a nearest neighbor search on data points from a standard dataset.

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